

Assignment 2 - DV2646

General Instructions

- This is an **individual assignment**. All submitted work must be your own.
- To pass the assignment, **you must successfully complete both tasks**.
- **Plagiarism in any form is strictly prohibited**.
- The assignment is graded as **G–U (Pass or Fail)**.
- All sections specified in the report requirements must be completed.
- The report must be written in a **single-column format**.
- Each major task must be clearly structured and labeled.

Submission Requirements

- Submit a **single ZIP file** via Canvas.
- The ZIP file must contain:
 - One PDF report
 - All implementation files (source code)
- **Do not include the dataset**.
- ZIP naming format:

`firstname_lastname.zip`

Question 1: LSTM Autoencoder-Based Anomaly Detection

Problem Description

In this task, you will design and implement **two different LSTM-based autoencoder models** for anomaly detection in sequential or structured data.

Anomalies are defined as **patterns that deviate significantly from normal behavior**, such as system faults, unusual activity, or rare events.

Dataset

You may choose **one dataset** from the following:

- NASA SMAP & MSL datasets
- Network Intrusion Detection dataset
- Financial Transaction dataset

You may optionally experiment with multiple datasets.

Tasks

- a) Preprocess the dataset:
 - Handle missing values
 - Normalize or scale features
 - Prepare sequences (for LSTM input)
- b) Design and implement **two different LSTM autoencoder architectures**:
 - Different depth, hidden sizes, or structure
 - Clearly justify architectural choices
- c) Train both models using reconstruction-based learning.
- d) Detect anomalies using reconstruction error:
 - Define a threshold
 - Justify threshold selection method
- e) Evaluate both models using:
 - Reconstruction error
 - Precision, Recall, F1-score
 - ROC-AUC (if applicable)
- f) Visualize:
 - Reconstruction error over time
 - Detected anomalies

Question 2: Baseline Model Comparison

Task

Select and implement **one baseline model** for anomaly detection.

Possible choices include:

- K-Means clustering
- DBSCAN
- Isolation Forest
- One-Class SVM
- Any well-justified alternative

Requirements

- Train and evaluate the baseline on the same dataset
- Use the same evaluation metrics as in Question 1
- Provide a fair comparison with LSTM models

Report Requirements

Your report must include the following sections:

1. Data Preprocessing

- Description of dataset
- Preprocessing steps and justification

2. Model Design

- Architecture details of both LSTM autoencoders
- Explanation of design choices

3. Training Configuration

- Loss function
- Optimizer
- Hyperparameters (learning rate, batch size, epochs)

4. Anomaly Detection Strategy

- Reconstruction error definition
- Threshold selection method

5. Evaluation

- Metrics: Precision, Recall, F1-score, ROC-AUC
- Quantitative results

6. Visualization

- Reconstruction error plots
- Detected anomalies visualization

7. Model Comparison

- Compare:
 - LSTM Model 1 vs Model 2
 - LSTM models vs baseline
- Discuss strengths and weaknesses

8. Discussion

- Model limitations
- Real-world applicability

9. Reproducibility

- Clear instructions to reproduce results

Code Requirements

- Code must be:
 - Readable
 - Well-structured
 - Fully executable
- All reported results must be reproducible.

Submission Deadline

Submission deadline: **15th May 2025, 23:59.**